Dataset Paper

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1 Abstract

There is a lack of slang datasets, especially non-english, that are well formatted and can be used for LLM research. To that end, I have collected and formatted slang terms from both Russian and Chinese datasets that have definitions in english as well as example usages in the host language and translations into english.

2 Introduction

Slang is a dynamic and culturally embedded aspect of language that evolves rapidly, often diverging significantly from formal linguistic norms. While recent advances in natural language processing (NLP) and large language models (LLMs) have improved text generation and understanding, these models struggle with slang due to its informal, context-dependent nature. This issue is further exacerbated by the lack of comprehensive slang datasets, particularly for languages other than English.

Existing efforts in computational slang modeling have primarily focused on English, with limited datasets available for other languages. Notably, an Indonesian slang dataset exists, but few resources cover Russian or Mandarin — two of the world's most widely spoken languages with rich and evolving slang lexicons. Without robust datasets for these languages, LLMs remain ill-equipped to understand and generate slang in multilingual contexts, limiting their ability to engage in natural, culturally relevant conversations. Efforts such as IndoCollex [Wibowo et al., 2021] for word level literal-to-slang translations for Indonesian and the subset of Urban Dictionary from [Ni et al., 2017] have brought forward a number of resources, but is it still an underdeveloped pool to pull from.

This whitepaper introduces a dataset of Russian and Mandarin slang as a foundational step toward improving crosslinguistic slang modeling in LLMs. By systematically collecting, annotating, and analyzing slang expressions in these languages, this dataset provides a critical resource for linguistic research and NLP applications. Our work aims to bridge the gap in non-English slang corpora, enabling more effective multilingual slang processing and contributing to the broader goal of making LLMs more culturally and contextually aware.

3 Collection Methodology

3.1 Mandarin Chinese

For the Mandarin dataset, I accessed this website: `https://www.chinese-tools.com/chinese/slang/`, which contains terms in Mandarin with definitions in English, example usage sentence in Mandarin, and a translation of that example sentence in English. The way the website was set up demanded that I scrape first for the terms and the links to their pages, which I did in "Collecting Words List.py". After each link was captured, I scraped each individually in "Words to Dataset.py". This resulted in a csv file with lines equating to a term, a definition, and zero or more example sentences. At this point, I took stats and did some additional cleaning in "Clean and Initial Stats.py", though these stats are not included as they are inaccurate to the final dataset. Here I checked to make sure each definition was in Chinese Characters and each example sentence and translation sentence was in Chinese and English respectively and that multiple definitions were grouped together(though still in their own line). Finally, in "Structuring Dataset.py" I read in the csv file using pandas and converted it to a DataFrame with the following format: term, definition [sentence: [example sentence, ...], translation: [translation sentence...]] This collected every term, definition, example sentence, and translation sentence into a csv file. I chose to avoid any additional translation steps to get the character-wise translation as it would have required the use of an additional resource and introduced additional complexity and variability into the dataset. Unfortunately, there was a number of terms lost in the cleaning process including all second definitions of terms with multiple definitions due to lack of standardization on the website.

3.2 Russian

For the Russian dataset, I accessed this website: `https://www.russki-mat.net/page.php?l=RuEn&a=%D0%`, which contained a number of Russian slang terms, english definitions, POS tags, example sentences, and translation sentences. This was done in the file "Russian Webscrape.py". I then did some cleaning with "Clean and Initial Stats.py" which were again discarded before cleaning some more and structuring with "Cleaning and Structuring.py". This left me with a CSV file of the same format as the example above.

In the definitions on the website, there were many occurrences of "form of [Russian word]" or "see [Russian word]," so many of the definition sentences still have Russian words in them. Our solution when using the dataset is to mask all instances of cyrillic with a [RUSS] token or something to that effect, but I have chosen to leave in the Russian in this dataset.

4 Statistics

For collecting the statistics, I simply ran the final dataset through the "collect stats.py" script. These are cleaned and reduced versions of the content of the website.

4.1 Mandarin Chinese

There are 643 total terms The average term length is 2.560 characters The average definition length is 8.506 words the average example sentence is 24.332 characters long the average translation is 14.518 words long

4.2 Russian

There are 1534 terms The average term length is 1.187 words The average definition length is 7.952 words the average example sentence is 17.235 characters long the average translation is 22.157 words long

5 Future Work

This dataset represents the initial step toward a more comprehensive approach to modeling slang in LLMs. Future work will expand upon this effort in several key directions:

5.1 Dataset Expansion and Multilingual Inclusion

Extending the dataset to include additional languages with rich slang traditions, particularly underrepresented languages with limited digital corpora. Continuously updating the dataset to capture slang evolution over time, incorporating new linguistic trends and regional variations.

5.2 Context-Aware Modeling

Developing methods to improve LLMs' ability to understand slang in context, distinguishing between different meanings based on situational factors. Exploring contrastive learning techniques to help models differentiate between formal and informal language usage.

5.3 Crosslinguistic Slang Representation

Investigating how slang translates across languages, identifying patterns of borrowing, adaptation, and code-mixing. Training multilingual models that can effectively map slang across different linguistic and cultural contexts without relying solely on direct translation.

5.4 LLM Fine-Tuning and Evaluation

Using the dataset to fine-tune existing LLMs, improving their slang comprehension and generation capabilities. Developing robust benchmarks to evaluate LLMs' performance on slang-heavy text, including measures for contextual accuracy and cultural appropriateness.

5.5 Application in NLP and AI Ethics

Integrating slang-aware models into chatbots, machine translation systems, and sentiment analysis tools. Addressing ethical concerns around slang processing, ensuring that models can differentiate between neutral, offensive, and reclaimed slang usage appropriately. By advancing these research directions, we aim to make slang modeling more linguistically inclusive, enabling LLMs to engage with informal language in a way that is both accurate and culturally nuanced.

References

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